Production methodology: A5

Companion to **Holistic Accuracy and the Limits of Lockean Derivation**, version 9, as adopted with two bibliography corrections. Prepared 30 October 2026; updated 31 October 2026.

The matching essay contains **5,597 words** under the conservative source count, including all non-bibliographic material. This anonymized report covers production, revision and verification; it carries forward the preceding production report and adds the readability revision described below.

Production setting and models

The essays were developed and revised during 21–31 October 2026 through two AI systems working with a shared project repository. This report reconstructs the production method from the retained drafts, review reports, source comparisons and production summary. It describes recorded work rather than treating a planned workflow as a completed experiment. Some historical details remain unverified.

The Claude-side records identify the scaffold designer, orchestrator, drafting agents and review agents as **Claude Opus 5.5**, with model string `claude-opus-5-5`; the orchestrator's environment was recorded as Claude Code 2.1.306 initially and 2.1.314 later in the run. These identifiers are reported from that system's records, not independently recovered from service telemetry. The other system was an **OpenAI Codex assistant**. A verified session-by-session model-version record for its earlier work was not available. It contributed candidate generation, critical review, mathematical checks, source comparison, substantive revisions and final document preparation. The preceding substantive finishing pass and the current readability revision also used Codex. Reviews by a third model (Grok), relayed by the operator on 31 October, and the Claude-side final audit and adoption on 31 October are described in the section "Final audit and adoption" below.

Candidate generation and selection

The initial AI-written scaffold proposed a broad funnel: survey the literature, generate candidates, reject anticipated or defective claims, draft survivors, obtain critical reviews, revise, audit citations and proofs, and package the results. Public research interests of the intended academic audience informed the literature areas and reviewer personas. The reviewer personas were composite subfield profiles given to models as prompts; none represented, imitated or involved any individual scholar. No scholar's participation or endorsement is claimed.

The original plan was reduced after budget and usage constraints. Three early essays were drafted from literature maps; prior-art checks led to replacement of some initial theses. A later problem survey listed 30 problems, shortlisted ten and selected six. Source availability helped determine which three were developed first. For these three, ideation agents produced fifteen proposals, five per topic. Recorded critical assessments rejected eight and repaired or merged seven into three essay plans. An additional Codex idea search generated 120 candidates and narrowed them through 40, 16 and eight; its survivors did not become these final essays. One other essay was withdrawn after its central objections remained unresolved.

Selection ultimately used the models' recommendations and the operator's choice among AI-produced outputs. The proposed formal draft tournament was not run. The process was exploratory and adaptive, not a preregistered evaluation. This matters because the final essays were selected from substantially more attempts than the three displayed here.

Instructions, scaffolding and revision

The recurring instruction pattern was to reconstruct the target's strongest argument, identify its assumptions, compare the closest primary literature, propose a precise increment, look for counterexamples, repair errors and narrow unsupported conclusions. Ideation tasks asked for multiple candidate arguments and an explicit account of what each would add. Critical tasks asked models to find prior art, invalid steps, misleading attributions and objections that would survive a charitable reading. Drafting tasks required a self-contained philosophical argument within the word limit. These are descriptions of the prompt functions, not a verbatim prompt archive.

Claude's workflow assigned separate roles for literature surveying, ideation, critical testing, drafting, revision, citation checking, formal checking and document checks. Some roles existed only in the planned scaffold; an editor stage and the proposed multi-round ranking tournament were not completed. Selected drafts received anonymous model-review packets containing the manuscript, a rubric and a main source. Some packets contained an additional comparator. The review roles included a subject specialist and an advocate of the position criticized. Their reports informed substantive revisions. Separate model instances and anonymous packets reduce some contextual influence but do not create human expert review or establish statistical independence between judgments.

Codex reviewed successive drafts against primary sources and supplied counterexamples, calculations and restrictions on claims. The two systems then incorporated each other's revisions. The last Claude pass added citation and proof-audit corrections. The preceding substantive edition incorporated those corrections and reviewed the changed claims. The current edition improves exposition and produces matching manuscripts and PDFs. It does not count a proposed check as completed or describe a previous draft's review as a new review of this version.

Research and computational tools

The systems used web search, author-posted papers, journal or repository copies, uploaded texts, text extraction and the shared repository. Bibliographic metadata checks supplemented reading; abstracts and metadata were not treated as substitutes for a full-text check of a theorem. Essential inaccessible texts were requested as documents rather than as human explanations.

Computational work used Python, exact rational arithmetic, symbolic algebra, finite-model enumeration, numerical integration and, in the Claude-side causation audit, an SMT solver. Such calculations checked examples and finite domains. Written proofs support claims quantified over all decision problems or all models in a stated class. Neither numerical agreement nor finite enumeration establishes originality. The essay-specific section below distinguishes checks performed during substantive review from the document checks repeated for this readability edition.

PDFs were generated from the final Markdown sources using Pandoc and XeLaTeX. The existing counting script counts title, abstract, headings, mathematical tokens, footnotes and appendices,

excluding only the bibliography. Text and metadata were checked for identifying strings, and rendered pages were visually inspected. The combined reading copy is an additional convenience document; the individual essay and its matching report are the paired documents.

Human involvement and context

The operator set the task and deadline, requested more candidates and stronger originality and significance, supplied or relayed source documents and model reviews, asked for further revision, and selected among AI-generated work. Recorded methodological suggestions included using relevant philosophical perspectives and demanding criticism. The available project review identified no substantive philosophical argument, example, distinction or repair supplied by the operator that entered these essays. This is a limited provenance finding, not a certification of every earlier session.

Claude's records state that automatic memory was disabled and personal-account retrieval was not used in its production workflow. A previous Codex disclosure recorded possible availability of unrelated personal context; the subsequent limited audit identified no human philosophical contribution but did not certify all historical context. That limitation remains disclosed. The preceding substantive pass did not retrieve personal context or use unrelated task histories to develop arguments. Raw conversations, human-input histories and internal run logs are not included in this report.

Resources and limitations of the record

The Claude-side usage tool, rerun on 31 October 2026 at 13:29 UTC, reports 3,571 requests to Claude Opus 5.5, including 3,173 subagent requests, across 86 transcript files covering 21 October 18:25 UTC to 31 October 13:29 UTC, and estimates approximately US$476.44 at list prices. The transcripts show activity in 72 ten-minute intervals, approximately twelve hours of occupied intervals across that period; this is not a measurement of processor time or summed agent time. These final figures cover the whole Claude-side run for all three essays, including the final audits, but not the last reconciliation of these reports. They are list-price estimates, not a bill: work used subscription access with usage limits. Reliable cumulative Codex token use, cost and machine time were unavailable. Total human time and a defensible cost allocation to this particular essay could not be established.

The earlier anonymous model reviews were critical; no calibrated human-expert assessment of these final versions has been completed. The workflow cannot demonstrate that the literature search was exhaustive, that model judgments are independent, or that the surviving contributions are important enough for publication. The most consequential remaining questions are stated below. No contest submission has been made as part of the preceding substantive pass.

Essay-specific development and checks

The initial proposal targeted the derivation of Lockean constraints from holistic accuracy and introduced a conjunction-sensitive record premium. Critical review corrected a closure claim; later revisions repaired consistency, ties and the bridge between maximization and rational permissibility. A model referee's refinement objection led to the cross-space discussion. Subsequent Codex review replaced an overstrong permission principle with Non-vacuity and restricted refinement

invariance to a sufficient route to the result. The family with true-belief value 1 and false-belief value $-n$ shows why fixed-space Lockeanism does not imply that cross-space assumption.

The published Dorst 2019 discussion and proofs of Propositions 4–5 were compared with the counterexample. The uploaded July 2021 penultimate author draft of Dorst and Mandelkern's *Good Guesses* was read, including its scoring rule, Filtering, belief application and Optionality argument. The published article was not substituted for that draft without a text comparison. The draft already produces the essay's two displayed belief sets: an exact calculation with the single Jamesian parameter $J = 125/64$ confirms this. The essay accordingly claims a more restricted increment about strict marginal truth-directedness and the characterization of maximizing cores, not the invention of utility-based guessing.

Earlier exact enumeration of all 256 three-world belief states gave the principal record-score optima 4.30 and 4.94 at the two stated credence functions. These checks were retained, not rerun for the last wording changes. Claude's subsequent audit supplied the qualifications that the relevant earlier utility theory is singular and that the null-world splitting discussion uses uniform values. This edition preserves those corrections. The mathematical content of this essay is unchanged from Claude's final manuscript at the reviewed checkpoint.

The remaining issue is interpretive and philosophical: whether Dorst's intended marginal-value restriction excludes the score, and whether the surviving optimizer result adds enough beyond the guessing literature. The stronger reading is explicitly discussed, rather than being represented as refuted. A final comparison with the published version of *Good Guesses* could still reveal wording or exposition differences from the inspected draft.

Readability revision

The revision explains the fixed-threshold question before introducing the formal setting, walks through the two credence distributions in separate paragraphs, distinguishes the sign of marginal value from its size, and connects the core-selection and refinement results to the philosophical issue. The formal statements, equations and proof text are unchanged; proof paragraphs were divided without changing their wording. Manuscript comparisons confirmed preservation of the formal content, and word counts, anonymity checks and PDF layout checks were repeated. The final review described next followed this prose revision; earlier substantive checks are identified above.

Final review

Codex re-read Dorst's published holistic derivation and Proposition 5 proof, and checked the added exposition against the unchanged formal statements. A targeted literature check identified Skipper's 2023 critique of refinement sensitivity in guessing. The essay now acknowledges that predecessor and distinguishes its comparison of fixed guesses from the unrestricted coarsening condition used here. The Holguín reference was corrected to the issue number printed on the journal PDF. No theorem or proof changed. The published version of Good Guesses remained inaccessible; the previously inspected July 2021 author draft remains the basis of that comparison. This was further review by an AI system involved in developing the essays, not independent expert endorsement.

Submission pass

The submission pass re-read the complete v9 manuscript, including all four proofs, and revisited Dorst's holistic derivation and Proposition 5 proof. No further manuscript change was required. The assessment retains the distinction between the unrestricted claim and the stronger interpretation that assumes restrictions on marginal values.

Final audit and adoption (31 October 2026)

Before adoption, a Claude-side citation, formal and compliance delta audit compared version 9 with the previously audited version 8. It found fourteen changed or added prose passages; one new reference (Skipper 2023), verified against registry metadata and the full open-access text; no new quotations; three new attributions to Skipper, all accurate; and theorem statements, definitions, equations, proofs and appendix identical to version 8 apart from whitespace and paragraph breaks. It found one defect: the corrected Holguín entry gave pages 1–34, but the article has 25 pages. The Claude orchestrator applied two bibliography patches verbatim: the Holguín page range became 1–25, and the issue number (7) was added to the Skipper entry. An optional wording patch was not applied. The bibliography is excluded from the word count, which remains 5,536 words. Injection and anonymity checks showed no hard flags.

On 31 October the operator relayed reviews of all three essays by a third model (Grok), saved verbatim. The Claude-side record of them notes that they reported no false lemma, no misrepresented source passage and no prior work already establishing the claimed contribution; their requests concern framing and exposition. The review of this essay recommended revise and resubmit. The delta audit mapped each request against version 9: most were already addressed; three presentational requests (why a holist should value the truth of the core rather than another interaction term, frameworks that do not guarantee a rational state at every credence, and intuition for the interval-length formula) remain only partly addressed. No change was made in response to the review beyond what version 9 already contained.

Final document pairing

The adopted essay is version 9 (`A5-Holistic-Accuracy-v9.md`) with the two bibliography corrections described above; the anonymous essay PDF was rebuilt from the corrected text. This report is paired with that text.

Late review edit (31 Oct)

An outside AI model (DeepSeek) wrote referee reports on the final essays, relayed unedited by the operator. A Claude-side substance-only triage found one real gap: the claim about Dorst's state-indexed family of measures was asserted without proof. The triage agent supplied a one-paragraph proof that the dilemma's second horn extends to such families; a check script confirmed it on finite cases, and the paragraph was replaced. No theorem statement or example changed. Final length: 5,597 words.